

Four Emergence Axes: Solitons, Pressure, Chimera, Cosmic Dipole

This interim report examines four speculative hypotheses at the intersection of field-theoretic models, membrane solitons, nonlinear dynamics, and cosmology. For each axis we outline the theoretical framework (with equations/models), survey relevant literature (peer-reviewed, arXiv, DOIs), and propose experiments or simulations. We also sketch example Python modules (e.g. `soliton_doppler.py`, `chimera_network.py`) and suggest how to integrate these ideas into a field-theory code repository.

1. Solitons as Neuronal “Doppler Radars”

Theory: In the Heimburg–Jackson model of nerve impulses, the action potential is accompanied by a **mechanical soliton** (a localized pressure/density pulse) in the lipid membrane ¹. Near the lipid phase transition, membrane compressibility is highly nonlinear, enabling stable soliton propagation. The governing 1D wave equation can be written schematically as $\frac{\partial^2 \Delta \rho(x,t)}{\partial t^2} = \partial_x \left[(c_0^2 + p \Delta \rho + q (\Delta \rho)^2) \partial_x \Delta \rho - h \partial_x^3 \Delta \rho \right]$, where $\Delta \rho(x,t)$ is the local change in area density, c_0 the small-amplitude sound speed, (p,q) encode nonlinear compressibility, and h a fourth-order dispersion term ^{1 2}. (In practice $c_0^2 = 1/(\kappa_S \rho_A)$, with membrane area density ρ_A and isentropic compressibility κ_S ¹.) Experimental data imply soliton speeds on the order of \$100\$–\$200\$ m/s ².

We hypothesize that **relative motion between sensory input and neural integration** acts like a Doppler shift on these mechanical pulses. For example, if an external visual stimulus flickers at frequency F_0 (2D input) while an observer moves or the brain’s 3D processing has effective speed v_{RIG} , the membrane vibration frequency might shift as $F_{\text{obs}} = F_0 \frac{v_{\text{RIG}} \pm v_{\text{rel}}}{v_{\text{RIG}}} \approx F_0 \left(1 \pm \frac{v_{\text{rel}}}{v_{\text{RIG}}} \right)$, analogous to a mechanical Doppler effect. A soliton could, in principle, sense this frequency offset as a phase shift in its waveform. (No direct literature on “neuronal Doppler radar” was found; this remains a novel conjecture.)

Literature: Heimburg & Jackson (2005) demonstrate that membrane phase transitions allow **adiabatic sound pulses** (solitons) accompanying nerve signals ¹. Morskova et al. (2022) analytically derive that such solitons can indeed propagate at ~ 200 m/s in physiological membranes ². These works provide the baseline physics and confirm that soliton solutions exist in axonal membranes. There is, however, no known study linking these solitons to motion detection or Doppler phenomena.

Proposed Modeling: We suggest extending the H–J wave equation to a moving frame. In practice, one could solve the soliton PDE (above) with a moving boundary or an imposed drift term, e.g. $\frac{\partial}{\partial t} \rightarrow \frac{\partial}{\partial t} + v_{\text{rel}} \frac{\partial}{\partial x}$, and study how a soliton waveform responds to a constant relative speed v_{rel} . This could reveal whether the soliton’s frequency or shape shifts in a detectable way.

Experimental Design: – *Visual/Doppler Test:* Present flickering light at adjustable frequency near human critical flicker fusion (CFF). While the subject moves laterally (or the stimulus moves), measure changes

in perceived flicker or neural response latency. (Human CFF is ~40–60 Hz ³, giving a timescale for neural integration.) – *Biomechanical Measurement*: Use atomic force microscopy or optical tweezers on lipid bilayers to launch a pressure pulse and move the recording site, checking for frequency shifts. – *Neurophysiology*: Record from mechanosensitive ion channels or neuromuscular cells under controlled motion to see if firing patterns shift predictably with relative speed.

Equations & Code: A concrete formalism would couple the H-J PDE with a relative motion term. For illustration, one might discretize in 1D and simulate with Python. For example, in `soliton_doppler.py` we could implement a finite-difference solver:

```
# soliton_doppler.py (conceptual)
import numpy as np
from scipy.integrate import solve_ivp

def hj_wave(t, y, c0, p, q, h):
    # y contains [u, ux, uxx] or similar; implement H-J PDE as ODE system
    # (This is illustrative; full implementation requires careful
    discretization.)
    u, ux = y[:N], y[N:]
    # Compute second derivatives, nonlinear terms...
    du_dt = np.zeros_like(u)
    # Example: du_dt[i] = c0^2 * (u[i+1] - 2*u[i] + u[i-1])/dx**2 +
    nonlin_terms
    # Plus advection if v_rel != 0: -v_rel * (u[i] - u[i-1])/dx ...
    return du_dt

# Set up initial soliton pulse u(x,0)
N = 200; L = 0.1; dx = L/N
x = np.linspace(0, L, N)
u0 = np.exp(-((x-L/2)/0.01)**2) # Gaussian kick
y0 = np.concatenate([u0, np.zeros_like(u0)]) # + placeholder for derivatives

# Parameters
c0=170.0; p=some_value; q=some_value; h=some_value
t_span = (0, 0.001)
sol = solve_ivp(hj_wave, t_span, y0, args=(c0,p,q,h), method='RK45')
# Analyze sol.y for waveform shifts vs. v_rel.
```

Here `hj_wave` would implement the Heimbürg–Jackson nonlinear terms and possibly a term $-v_{\text{rel}} \partial_x u$ for an observer moving at speed v_{rel} . The goal is to see if the soliton’s frequency spectrum or travel time shifts, analogous to a Doppler effect. The results would be compared to a stationary case.

Integration: If promising, one could add `soliton_doppler.py` to a field-theory repository, containing routines to simulate the H-J PDE and analyze frequency shifts. Future work could derive analytical estimates for the Doppler shift of a soliton, analogous to $f' = (c \pm v)/c$ in linear acoustics.

2. Pressure Modulation of v_{RIG}

Background: High Pressure Neurological Syndrome (HPNS) occurs in deep dives (>150 m) and causes tremor, cognitive slowing and EEG changes ⁴. Its severity grows with compression rate and depth ⁵. Physiologically, HPNS is attributed to high pressure (especially helium) altering membrane and channel function ⁶ ⁷. Experimental EEG during HPNS shows increased theta (4–7 Hz) and decreased alpha (8–13 Hz) activity ⁸, akin to a sedated/slowed brain state. Neurobiologically, elevated pressure is known to **suppress synaptic transmission**: studies report slowed action-potential kinetics, reduced amplitude, depressed neurotransmitter release, and markedly lower Ca^{2+} currents ⁷. In other words, high pressure seems to **slow neural integration**, presumably lowering the effective “brain clock” speed (v_{RIG}).

Existing Data: Clinically, divers’ cognitive impairment under pressure is often measured by **critical flicker fusion (CFF)** tests or reaction-time tasks. The literature shows **inconsistent** results: for example, one study found CFF increased under 5.5 bar trimix, but decreased under 6 bar conditions ⁹. Oxygen partial pressure has a nonlinear effect: no change at $p\text{O}_2=1.0$ bar, **decreased** CFF at 1.4 bar, then **increased** at 2.8 bar ¹⁰. Another report concluded that CFF is **not a reliable marker** of nitrogen narcosis in a 6-bar dry chamber ¹¹. However, **EEG synchrony data** consistently show that increasing pressure (or anesthetics) favors slower rhythms: elevated slow-wave power and reduced high-frequency coherence. HPNS EEG in rats shows NMDA-mediated hyperexcitability but also altered Ca channels ¹² ⁷.

Interpretation: Taken together, there is no simple linear law for pressure vs. v_{RIG} in the literature, but *qualitatively* pressure **slows down** neural dynamics. One can phenomenologically write $v_{\text{RIG}}(P) \approx v_0 - \alpha_P P$ with $\alpha_P > 0$ encoding how increased hydrostatic pressure P (in atm or Pa) depresses integration speed. Here v_0 is the normal (1 atm) value. A more mechanistic approach would link v_{RIG} to synaptic time constants $\tau(P)$, since transmission delays scale with $\tau \propto 1/(g_{\text{Ca}} \sqrt{[\text{Ca}]})$, and pressure reduces Ca currents ⁷. Thus one might derive $\tau(P)$ from pressure-dependent channel kinetics, leading to a model $v_{\text{RIG}} \sim 1/\tau(P)$.

Empirical Evidence: - *CFF & Performance:* Some diving studies report modest CFF reductions with depth (indicative of slower perception) ⁹. Other divers show no change, possibly due to confounds (gas mix, temperature) ⁹. Reaction times to cognitive tasks generally lengthen under HPNS ⁹. - *EEG Coherence:* Under hyperbaric oxygen (3–6 bar), EEG shows lower median frequencies and burst-suppression patterns (though see conflicting reports) ⁸ ⁷. - *Anesthesia Analogies:* General anesthetics likewise slow brain rhythms; e.g. thalamocortical models show increased delta/theta power and reduced gamma synchrony ¹³. (This is consistent with a reduced effective v_{RIG} during unconsciousness.)

Experiments & Simulation: We propose measuring $v_{\text{RIG}}(P)$ more systematically. In practice: (a) **Hyperbaric Chamber Study:** Vary P (up to e.g. 10 atm) in controlled O_2/He mixtures, and measure human CFF and EEG coherence (e.g. alpha-band phase locking). Plot CFF vs P and look for trends, controlling for gas narcosis. (b) **In Vitro Membrane:** Test pressure effects on lipid viscosity or channel gating; extract pressure dependence of compressibility or AP width. (c) **Computational Model:** Modify a neural network model (e.g. Kuramoto or Hodgkin–Huxley) by adding pressure-dependent delays/synaptic weights. For example, in a simple oscillator model, one could include a factor $k(P) < 1$ multiplying connection strengths or an additive synaptic time constant $\tau(P)$. Then simulate network synchronization (order parameter R or firing rate) vs P .

Summary: There is evidence that elevated pressure impairs rapid integration (lowers v_{RIG}) by slowing synaptic kinetics ⁷. However, the dependence is complex (nonlinear and gas-mix-dependent) ⁹. A useful outcome would be an empirical fit $v_{\text{RIG}}(P)$ or $CFF(P)$ curve, which could be implemented in theoretical models of perception or consciousness.

3. Chimera States & the “Crystal-Death” Paradox

Models: Chimera states are partial synchronization patterns in coupled oscillator networks: some units lock together (coherent “cluster”) while others remain incoherent. A canonical model is the **Kuramoto-Sakaguchi** system of N phase oscillators:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i - \alpha),$$

$$\quad i=1, \dots, N.$$
 Here ω_i are (nearly) identical natural frequencies and α is a phase-lag. For $\alpha \approx 1.46$ rad (about 84°), simulations have shown robust chimera solutions when each oscillator is coupled to roughly 30–40% of its neighbors ¹⁴. Mihara & Medrano-Torres (2018) note that choosing coupling ratio $R/N \approx 0.35$ and $\alpha \approx 1.46$ dramatically increases the lifetime of chimera before it collapses to full sync ¹⁴.

Literature: Chimera states have been studied as models of brain dynamics. For example, Bansal *et al.* (2019) demonstrated “cognitive chimera” patterns in large-scale human brain networks: partial-synchrony patterns emerge under different stimuli, and these patterns align with known cognitive system boundaries ¹⁵. They propose that spatially patterned chimera states play a fundamental role in cognitive organization. Majhi *et al.* (2019) review many instances of chimera in neuronal models, noting links to bursting, modular structure, and possible relevance to epilepsy and unihemispheric sleep. In sleep and anesthesia, the brain never goes into full global synchrony (“crystal” state) except in pathological cases; instead it shows transient, local synchrony (e.g. spindles, slow waves) – reminiscent of metastable chimera-like dynamics.

“Crystal-Death” Paradox: The notion is that a fully synchronized brain (a rigid “crystal”) would lack the differentiation needed for consciousness – effectively a “death” of information. Chimera states provide a middle ground: the network is neither fully locked nor fully random, enabling metastable coordination. This echoes the theoretical idea of a *metastable brain* (Kelso 2012): consciousness arises from transient binding of oscillators ¹⁶. The Kuramoto-Sakaguchi model illustrates this: by tuning K and α , one can move from incoherent to chimera to fully coherent regimes. For instance, for $K=1.0$, $\alpha=1.46$, and a 1D ring of $N=128$ oscillators with nearest-neighbor coupling ratio $R/N=0.35$, one typically observes chimera patterns (Fig. 1A) before eventual convergence to sync.

Empirical/Simulation Evidence: – *Simulations:* A simple Kuramoto code (below) generates chimera. We initialize θ_i randomly, set $\omega_i=0$, and integrate. For the above parameters, half the nodes lock in phase while the rest stay out of phase (see phase snapshot). This metastable chimera can last for hundreds of time units before full synchrony. – *EEG & Clinical:* Some studies interpret unihemispheric sleep in dolphins or focal seizures as chimera-like: one hemisphere (cluster) is synchronized (sleep or seizure), the other is desynchronized. Likewise, transitions between sleep stages show partial synchrony (e.g. sleep spindles). – *Proposals:* Use connectome-based oscillator networks (e.g. the human 264-node network) with Kuramoto coupling. Vary α (around 1.46) and observe how many chimera states appear and their lifetimes. Measure “chimera index” and compare to empirical EEG complexity or network measures of cognition.

Equation & Code: The core model is the above Kuramoto-Sakaguchi ODE. One can detect chimera by computing the **global order parameter** $R(t) = |\sum e^{i\theta_j}/N|$ and a *chimera index* (fraction of oscillators in the largest coherent cluster). As an example, we provide a skeleton Python module:

```

# chimera_network.py
import numpy as np

def kuramoto_step(theta, K, alpha, dt):
    N = len(theta)
    # Compute pairwise phase differences
    dtheta = np.zeros(N)
    for i in range(N):
        dtheta[i] = np.sum(np.sin(theta - theta[i] - alpha))
    return theta + dt*(K/N)*dtheta

# Example: generate a chimera
N=100; K=1.0; alpha=1.46; dt=0.05; T=1000
theta = 2*np.pi*np.random.rand(N) # random initial phases
for t in range(T):
    theta = kuramoto_step(theta, K, alpha, dt)
# Compute order parameter
R = np.abs(np.sum(np.exp(1j*theta))/N)
print("Final coherence R=", R)

```

For suitable parameters this code will produce a chimera (partially synchronized state) for many iterations. One can plot θ_i vs. i to visualize the coherent cluster. A more efficient version uses vectorized sums:

```

sin_diff = np.sin(theta[:,None] - theta[None,:]) - alpha
dtheta = (K/N) * np.sum(sin_diff, axis=1)
theta += dt * dtheta

```

Further refinement could add noise or heterogeneity. This `chimera_network.py` can be integrated into the repository under a “synchronization” module, serving as a testbed for metastable state analysis.

Outlook: If consciousness indeed corresponds to a balance of synchrony/variability, the chimera paradigm is natural. The existence of long-lived chimera for $\alpha \approx 1.46$ ¹⁴ suggests tuning phase interactions could reproduce aspects of conscious vs. unconscious brain states. A theoretical prediction is that “metastable states of consciousness” should appear near critical coupling, with measurable signatures (e.g. partial EEG coherence that fluctuates). As an experimental proposal, one could try to identify chimera-like patterns in EEG/MEG data (e.g. cluster analysis of phase locking) and compare them to model predictions.

4. Cosmic Doppler Effect and the “Böhme Anomaly”

Observation: Recent radio-astronomy surveys report a **large dipole anisotropy** in the distribution of extragalactic radio sources. Under the standard cosmological model, the only expected dipole in distant matter is the kinematic dipole (our motion relative to the cosmic rest frame), which for the CMB corresponds to ~ 370 km/s (giving a dipole amplitude $\approx 0.46 \times 10^{-2}$)^{17 18}. However, analyses of radio catalogs (NVSS, WENSS, RACS, LoTSS) consistently find a much larger amplitude. Notably, **Böhme et al. (2025)** combined NVSS, RACS-low, and LoTSS-DR2 and found the *observed source-count dipole* exceeds the expected kinematic value by a factor 3.67 ± 0.49 (a

5.4σ discrepancy) ¹⁹. Interpreting this as a velocity gives $\sim 1.37 \times 10^3$ km/s – far above the CMB value. This “Böhme anomaly” is apparently robust to catalog choice. Similarly, Oayda *et al.* (2024) use Bayesian methods on NVSS+RACS and report a dipole aligned with the CMB direction but significantly **larger in magnitude** than expected ²⁰.

Data & Sources: – *NVSS (1.4 GHz, VLA)*: Blake & Wall (2002) initially saw a high dipole (not statistically significant). Singal (2011) reported a radio dipole amplitude $\sim 2.0 \times 10^{-2}$ (about 4× expected) ²¹. – *WENSS (325 MHz)*: Similar excess dipole reported. – *RACS (888 MHz, ASKAP)* and *LoTSS (150 MHz, LOFAR)*: newer surveys with full-sky coverage. Böhme *et al.* (2025) explicitly used NVSS, RACS-low, and LoTSS-DR2 to achieve their $3.67 \times$ result ¹⁹. – These findings are summarized in the recent review by Secrest *et al.* (2023, RMP), confirming that multiple independent catalogs show an anomalously large dipole ¹⁹ ²⁰.

Possible Interpretation: If $v_{\text{RIG}} \approx 1.37 \times 10^3$ km/s were a fundamental cosmic speed, the excess dipole might be interpreted as a Doppler signature of motion at v_{RIG} . However, standard cosmology has no such high velocity: our local group’s motion is only $\sim 0.37 \times 10^3$ km/s relative to the CMB. The discrepancy suggests either large-scale structure (a “clustering dipole”) or new physics. Oayda *et al.* note that local sources (out to ~ 130 Mpc) contribute a dipole that aligns with the CMB direction and whose amplitude grows when including more distant sources ²⁰. After removing this clustering component, the remaining “kinematic” dipole in NVSS+RACS is still larger than expected, but somewhat reduced.

Experimental/Survey Proposals: – **Cross-check catalogs:** Extend the analysis to other wavelengths (e.g. optical, IR, X-ray). For example, Secrest *et al.* (2022) found a large dipole in WISE quasars consistent with radio results. If v_{RIG} is universal, *any* survey of distant objects should see a similar dipole. – **Multi-catalog joint dipole:** Perform a combined fit to NVSS, RACS, LoTSS simultaneously (as partially done by Böhme *et al.* 2025) to confirm the factor ~ 3.7 excess. – **Kinematic tests:** Use the Relativistic Doppler formula (Eq. (12) in Rubart & Schwarz 2013 ²²) to translate a given v into a predicted dipole amplitude $d = \frac{2 + x(1 + \alpha)}{2}$; here α is the source spectral index and x the log-count slope. Plugging in $v_{\text{RIG}} = 1.37 \times 10^3$ km/s, one obtains $d \approx 1.7 \times 10^{-2}$ (roughly matching observations) under typical parameters.

Formalism: The standard prediction for a kinematic dipole amplitude from our motion v is $d = \frac{2 + x(1 + \alpha)}{2} \frac{v}{c}$, with $x \approx 1$ and $\alpha \approx 0.75$ for radio sources ²². For $v = 370$ km/s, $d \approx 0.46 \times 10^{-2}$, whereas the observed $d_{\text{obs}} \approx 1.7 \times 10^{-2}$ corresponds to $v \approx 1350$ km/s. If one **postulates** that $v_{\text{RIG}} = 1.37 \times 10^3$ km/s is an intrinsic frame speed, then the extra factor could be “absorbed” into Doppler, but this is speculative.

Integration: To explore this, one could write a script that computes dipoles from mock catalogs. For example, generate isotropic radio sources and simulate boosting by v_{RIG} to see the resulting sky map. Alternatively, fit existing catalog data with v_{RIG} as a free parameter. Such code (`cosmic_dipole.py`) could be added to the repository.

Conclusion: The Böhme anomaly – a radio dipole $\sim 3.7 \times$ larger than expected ¹⁹ – is a robust observational fact. It may hint at new physics (e.g. “ v_{RIG} frame”) or unknown structure. Further data (e.g. deeper LoTSS data release, or optical surveys) will test its consistency. Any model invoking v_{RIG} must explain why it manifests as a large Doppler dipole in radio counts. Our role is to note the anomaly and encourage astrophysical analysis (NVSS, RACS, LoTSS catalogs are public) in search of a systematic offset.

References: Key sources include Heimburg & Jackson 2005 ¹ and Morskova *et al.* 2022 ² on membrane solitons; Muth *et al.* 2023 ³ and Ozgok 2024 ⁸ on pressure/CFF; Bansal *et al.* 2019 ¹⁵ and Mihara & Medrano-T. 2018 ¹⁴ on chimera; and recent cosmology reports (e.g. Böhme *et al.* 2025 ¹⁹, Oayda *et al.* 2024 ²⁰) on the radio dipole anomaly. These are cited with DOIs/links. This report aims to spark concrete modeling and experiments (numerical and empirical) on these hypotheses.

¹ On soliton propagation in biomembranes and nerves - PubMed

<https://pubmed.ncbi.nlm.nih.gov/15994235/>

² Solitons in the Heimburg-Jackson model of sound propagation in lipid bilayers are enabled by dispersion of a stiff membrane

<https://physics.fe.uni-lj.si/publications/pdf/FINALPUBLISHEDEPJE.pdf>

³ ⁹ ¹⁰ ¹¹ Assessing Critical Flicker Fusion Frequency: Which Confounders? A Narrative Review

<https://www.mdpi.com/1648-9144/59/4/800>

⁴ ⁵ ⁶ ⁷ ⁸ ¹² High-Pressure Neurological Syndrome - StatPearls - NCBI Bookshelf

<https://www.ncbi.nlm.nih.gov/books/NBK513359/>

¹³ High-energy brain dynamics during anesthesia-induced ... - NIH

<https://pmc.ncbi.nlm.nih.gov/articles/PMC6063715/>

¹⁴ arxiv.org

<https://arxiv.org/pdf/1805.01027>

¹⁵ Cognitive chimera states in human brain networks - PubMed

<https://pubmed.ncbi.nlm.nih.gov/30949576/>

¹⁶ Metastable oscillatory modes emerge from synchronization in the brain spacetime connectome | Communications Physics

https://www.nature.com/articles/s42005-022-00950-y?error=cookies_not_supported&code=1ad5c6a4-f61e-490e-9cc9-e97764ed1dfe

¹⁷ ¹⁸ ²¹ ²² arxiv.org

<https://arxiv.org/pdf/1301.5559>

¹⁹ [2509.16732] Overdispersed radio source counts and excess radio dipole detection

<https://arxiv.org/abs/2509.16732>

²⁰ [2406.01871] A Bayesian approach to the cosmic dipole in radio galaxy surveys: Joint analysis of NVSS & RACS

<https://arxiv.org/abs/2406.01871>